

The `banjofret` package

Colour-coded fretboard diagrams for the 5-string banjo

banjofret contributors

Version 1.0.0 — 2026/09/28

Abstract

`banjofret` typesets a clean, black-and-white diagram of a 5-string banjo fretboard (24 frets, short fifth string starting at fret 5) with the chromatic note name printed on every string/fret position. You then *paint* the notes you care about — single positions such as (3,5), whole strings, whole frets, or every occurrence of a given note — using a small palette of minimalist colours chosen to support learning. Note labels are always drawn on top of the strings and of the colours. It is built on `TikZ` and `expl3` and works with `pdfLATEX`, `XeLATEX` and `LuaLATEX`.

Contents

1	Introduction	1
1.1	Requirements	1
1.2	Installation	1
2	Quick start	1
3	How the diagram is organised	4
4	Painting commands	4
5	Chords, scales and arpeggios	6
6	Colours	7
7	Options	8
8	Tips and troubleshooting	9
9	Roadmap	9
10	Implementation	9
10.1	Source code	9
10.1.1	Package identification and dependencies	9
10.1.2	Messages	10
10.1.3	Colour palette	10
10.1.4	Variables and constants	10
10.1.5	Options	11
10.1.6	Note names and pitch classes	11
10.1.7	Colour resolution	12
10.1.8	Painting commands	12
10.1.9	Geometry helpers	14
10.1.10	Drawing the fretboard	15
10.1.11	The <code>banjofret</code> environment	18

1 Introduction

The goal of the package is to let you design *your own* banjo diagrams — scales, chords, arpeggios, licks, positions — with as little typing as possible. There is one template (the fretboard) and a handful of commands to paint it:

- you always get the full chromatic fretboard, open-G tuned, in black and white;
- you paint by position, by string, by fret or by note name;
- colours are short names (`blue`, `root`, `amber`, ...);
- note names are drawn last, so they stay legible over strings and colours.

1.1 Requirements

L^AT_EX 2022-06-01 or newer (for the built-in `\NewDocumentCommand`), plus the packages `tikz`, `xcolor` and `l3keys2e`, all part of every standard T_EX Live and MiK_TE_X installation. The package loads `tikz` (with the `calc` library) and `xcolor` itself.

1.2 Installation

Run `latex banjofret.ins` to generate `banjofret.sty`, and place it (together with this PDF) in your local `texmf` tree, for instance `TEXMF/tex/latex/banjofret/`. For a one-off project it is enough to keep `banjofret.sty` next to your `.tex` file. On Overleaf, upload `banjofret.sty` to the project.

2 Quick start

```
\documentclass{article}
\usepackage[landscape,margin=1.5cm]{geometry}
\usepackage{banjofret}
\begin{document}

\begin{banjofret}[title={My first diagram}]
  \bfpaint{(3,5)}{blue}           % string 3, fret 5
  \bfpaintnote{G, B, D}{root}     % every G, B and D on the neck
\end{banjofret}

\end{document}
```

With no painting commands at all you get the empty template with all 24 frets. It is wide, so landscape pages (or a 12-fret window, see Section 7) are recommended:

```
\begin{banjofret}[title={Chromatic notes -- 5-string banjo, open-G tuning}]
\end{banjofret}
```

Chromatic notes – 5-string banjo, open-G tuning

1	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D
2	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B
3	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G
4	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D
5					G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	D $\sharp$ E $\flat$	E	F	F $\sharp$ G $\flat$	G	G $\sharp$ A $\flat$	A	A $\sharp$ B $\flat$	B	C	C $\sharp$ D $\flat$	D	
		1	2	3	4	5 •	6	7 •	8	9	10 •	11	12 ••	13	14	15 •	16	17 •	18	19 •	20	21	22 •	23	24 ••

3 How the diagram is organised

Strings. Strings are numbered as on a banjo: string 1 is the thinnest of the four full-length strings (high D) and string 4 the thickest (low D). String 5 is the short drone string. By default string 1 is drawn on top and string 5 at the bottom, like in banjo tablature (`string order=tab`); use `string order=reverse` for the opposite.

Frets. Fret 0 is the open string and is drawn in the column left of the nut (the thick vertical bar). Frets 1 to 24 follow. Frets have equal width: the diagram is a schematic and not a scale drawing.

The fifth string. The fifth string starts at fret 5, where its tuning peg sits. The disc at position (5,5) is the open fifth string (G); (5,6) is $G\sharp/A\flat$, and so on. Positions (5,0) to (5,4) do not exist and are never drawn.

Position notation. A position is written `(string,fret)`. Thus (3,5) means string 3 at fret 5. Several positions are separated by *semicolons*: (3,5); (2,1); (1,0).

Default tuning. The default is standard open-G (G4–D3–G3–B3–D4):

String	1	2	3	4	5
Note	D	B	G	D	G

Other tunings can be selected with the `tuning` option (see below).

4 Painting commands

All painting commands must be used *inside* a `banjofret` environment. If two commands paint the same position, the *last* one wins, so you can first paint a broad set of notes and then override individual positions. The last argument of every command is a colour (Section 6).

`banjofret`
(*env.*)

```
\begin{banjofret}[\langle options \rangle] \langle painting commands \rangle \end{banjofret}
```

Typeset one fretboard diagram. The optional argument takes the options of Section 7.

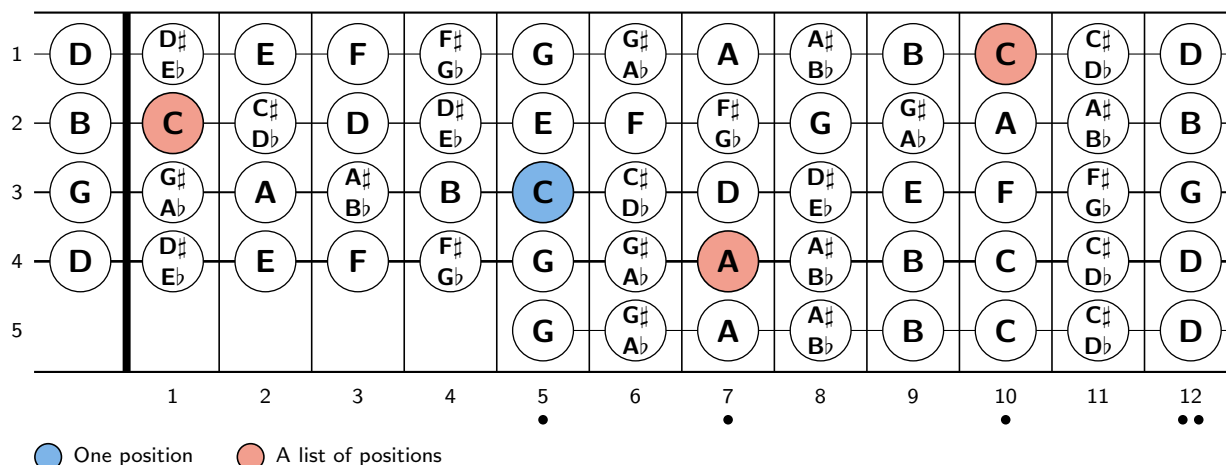
`\bfpaint`

```
\bfpaint{\langle positions \rangle}{\langle colour \rangle}
```

Paint one or more `(string,fret)` positions. Parentheses are optional; positions are separated by semicolons.

```
\begin{banjofret}[last fret=12, title={Painting single positions}]
\bfpaint{(3,5)}{blue}
\bfpaint{(2,1); (4,7); (1,10)}{coral}
\bflegend{blue=One position, coral=A list of positions}
\end{banjofret}
```

Painting single positions

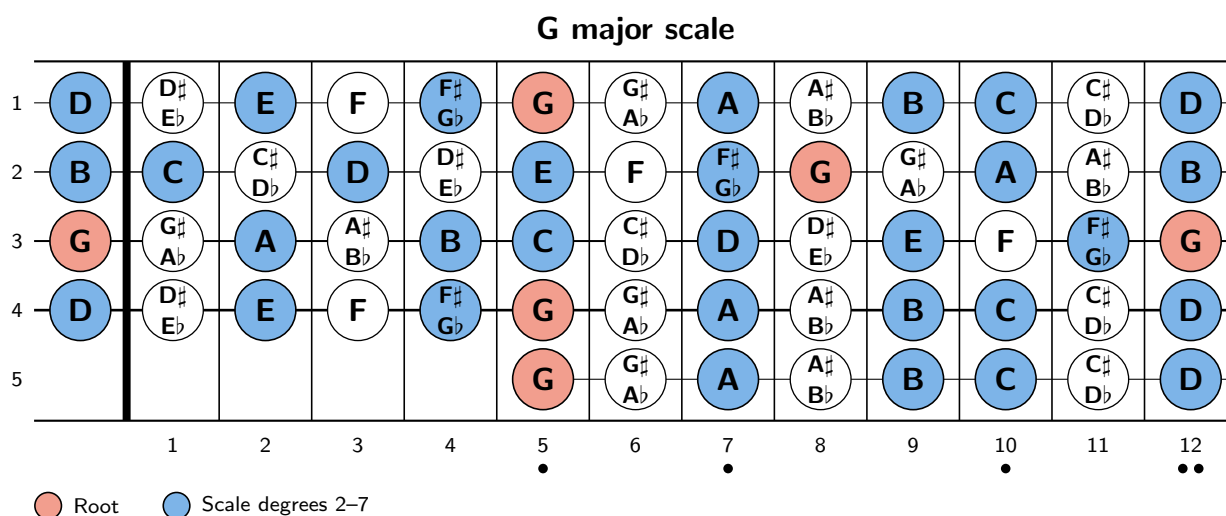


`\bfpaintnote`

`\bfpaintnote{<notes>}{<colour>}`

Paint *every* occurrence of one or more notes on the neck. Notes are comma-separated names: a letter A–G followed by an optional accidental, which can be written **s** or **#** for sharp and **b** for flat: C, **Fs**, **F#**, **Bb**. Enharmonic spellings are equivalent (**Fs** and **Gb** paint the same discs). This is the fastest way to draw scales and arpeggios:

```
\begin{banjofret}[last fret=12, title={G major scale}]
  \bfpaintnote{A, B, C, D, E, Fs}{blue}
  \bfpaintnote{G}{root}
  \bfllegend{root=Root, blue=Scale degrees 2--7}
\end{banjofret}
```



`\bfpaintstring`

`\bfpaintstring{<string>}{<frets>}{<colour>}`

Paint a run of frets on one string. The fret range is either a single fret (5) or a range (2–7).

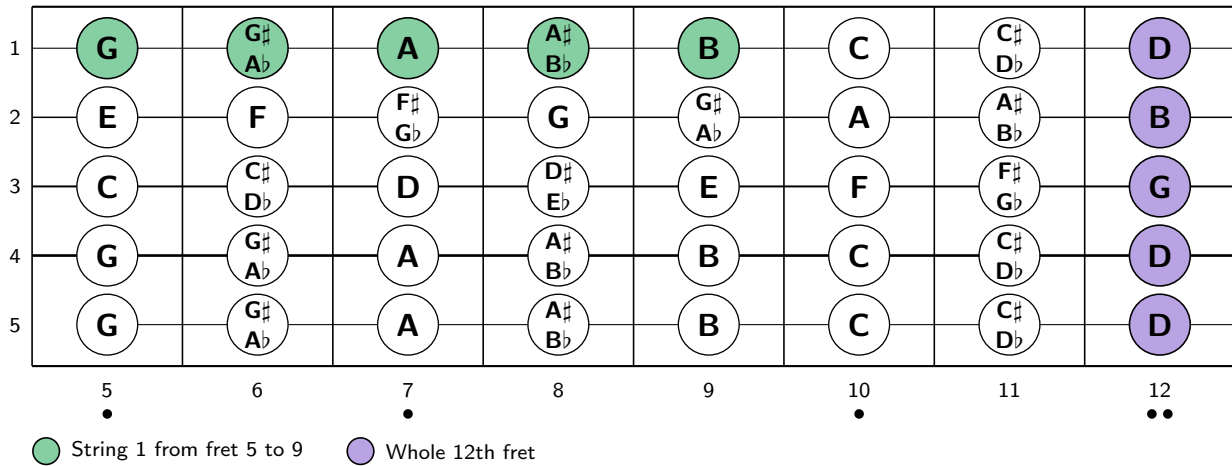
`\bfpaintfret`

`\bfpaintfret{<fret range>}{<colour>}`

Paint the same fret, or a range of frets, on all strings.

```
\begin{banjofret}[first fret=5, last fret=12, title={Ranges: one string, one fret}]
  \bfpaintstring{1}{5-9}{green}
  \bfpaintfret{12}{violet}
  \bfllegend{green=String 1 from fret 5 to 9, violet=Whole 12th fret}
\end{banjofret}
```

Ranges: one string, one fret



`\bflegend`

`\bflegend{<colour>=<label>, ...}`

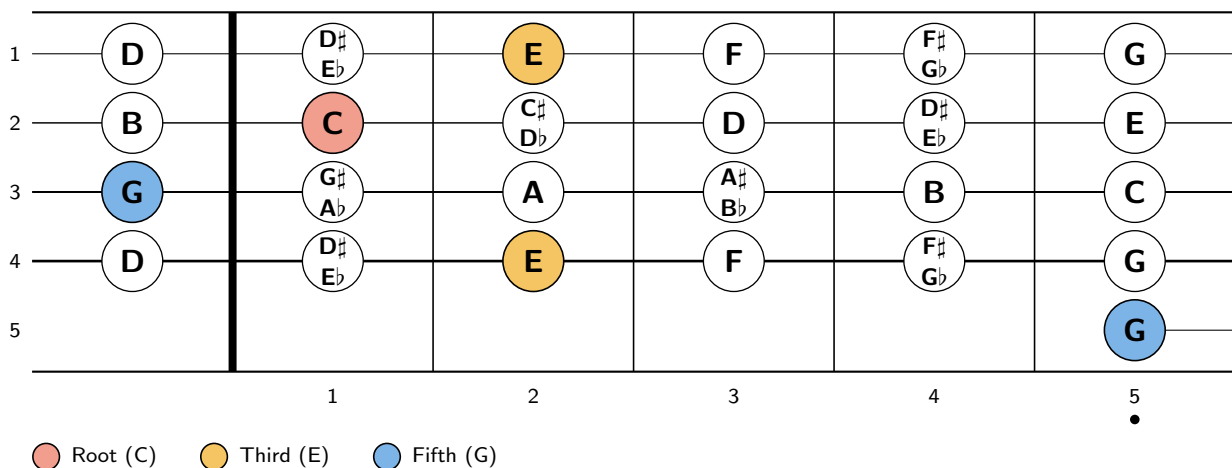
Add a legend line below the diagram. Use braces around labels that contain commas or equal signs.

5 Chords, scales and arpeggios

Chord shapes are most readable as position lists with the chord-tone colours `root`, `third`, `fifth` and `seventh`:

```
\begin{banjofret}[last fret=5, title={C major chord (open-G tuning)}]
  \bfpaint{(2,1)}{root}
  \bfpaint{(4,2); (1,2)}{third}
  \bfpaint{(3,0); (5,5)}{fifth}
  \bflegend{root=Root (C), third=Third (E), fifth=Fifth (G)}
\end{banjofret}
```

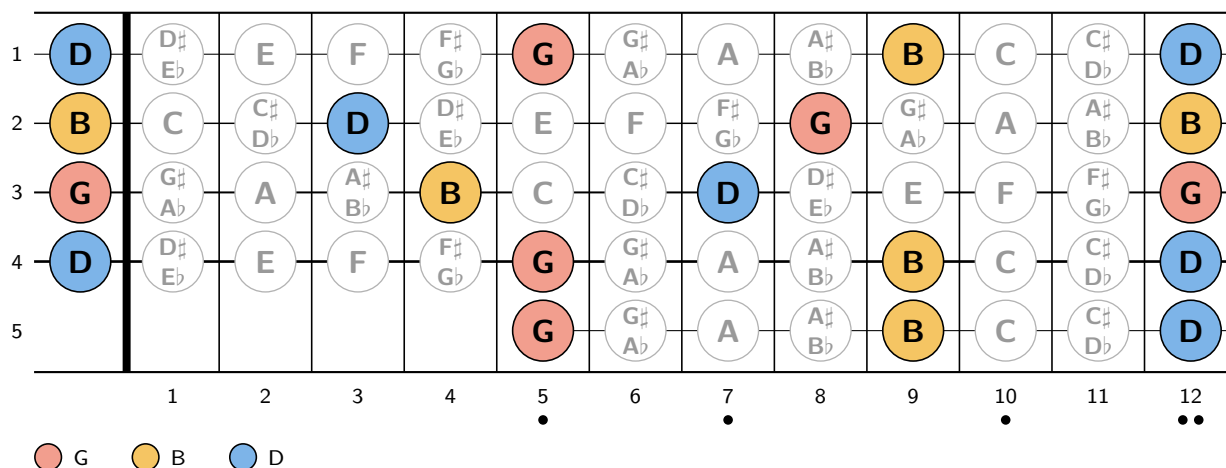
C major chord (open-G tuning)



Whole-neck views work best with `focus`, which greys out every note that is not painted, so the eye finds the target notes at once:

```
\begin{banjofret}[last fret=12, focus, title={G major triad -- focus mode}]
  \bfpaintnote{G}{root}
  \bfpaintnote{B}{third}
  \bfpaintnote{D}{fifth}
  \bflegend{root=G, third=B, fifth=D}
\end{banjofret}
```

G major triad – focus mode



6 Colours

The colour argument of every painting command can be one of the palette names below. The names are resolved to the package colours `bf<name>` (for instance `blue` $\rightarrow$ `bfblue`), which can therefore also be used with any standard `xcolor` command. Any other `xcolor` expression (`red!30`, a colour defined with `\definecolor, \dots`) is accepted as it is.

Name	Swatch	HTML	Suggested meaning
blue		7DB4E8	calm, main information
green		86CFA3	targets, resolution, “safe” notes
amber		F5C563	attention, passing emphasis
coral		F29E8E	the root, the most important note
violet		B9A3E3	colour tones, extensions
teal		7BCFCF	second family of tones
gray		C9CED6	neutral, context, ghost notes

Semantic aliases make diagrams self-consistent across a whole book:

Alias	=	Alias	=
root	coral	seventh	violet
third	amber	extension	teal
fifth	blue	passing	gray
target	green		

Design rationale. The palette follows a few well-established guidelines from the literature on multimedia learning and colour accessibility (they are guidelines, not laws): (1) a *small* set of hues, so that the meaning of each colour can be remembered; (2) one *consistent* meaning per colour — the warm coral is kept for the most salient item, the root, while cool colours carry supporting information; (3) soft, mid-light tones with black text, which keeps contrast high and the page calm, and avoids the visual vibration of pure primaries; (4) hues that stay distinguishable under the common forms of colour-vision deficiency; and (5) redundant coding: the note *name* is always printed inside the disc and a legend can be added, so colour is never the only channel.

Your own colours. Define them with `xcolor` and use them by name:

```

\definecolor{mine}{HTML}{FFB3C1}
\bfpaint{(3,5)}{mine}

```

7 Options

Options are given in the optional argument of the environment. To change a default for the whole document, pass them as package options, for example `\usepackage[last fret=12]{banjofret}`, or set them in the preamble with:

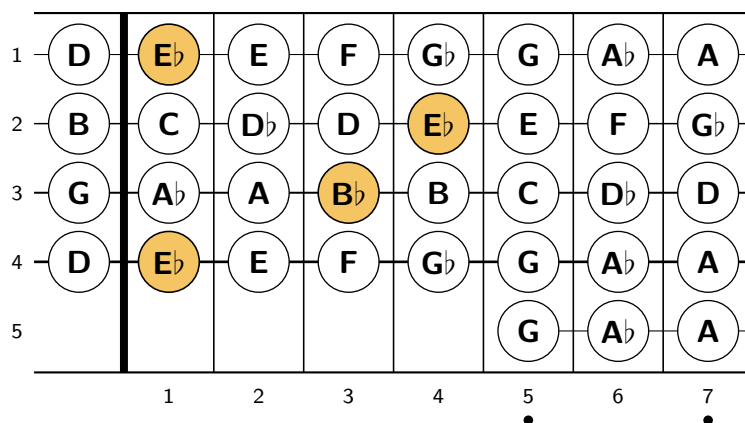
`\banjofretset`

`\banjofretset{<options>}`

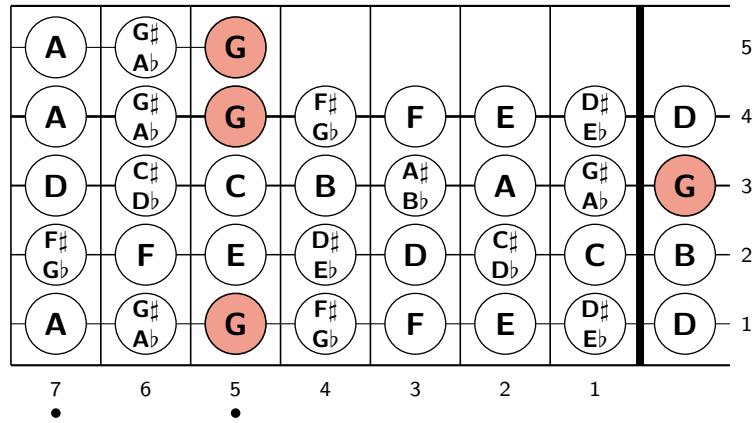
Option	Default	Meaning
<code>first fret</code>	0	First fret shown. 0 shows the open-string column and the nut.
<code>last fret</code>	24	Last fret shown (maximum 24).
<code>width</code>	<code>\linewidth</code>	Total width of the diagram; any T _E X length.
<code>accidentals</code>	<code>both</code>	<code>sharp</code> (F $\sharp$), <code>flat</code> (G $\flat$) or <code>both</code> (stacked, F $\sharp$ over G $\flat$).
<code>tuning</code>	D,B,G,D,G	Five notes, string 1 to string 5. Write accidentals with s/b, e.g. D,C,G,C,G.
<code>string order</code>	<code>tab</code>	<code>tab</code> : string 1 on top. <code>reverse</code> : string 5 on top.
<code>left handed</code>	<code>false</code>	Mirror the diagram (nut on the right).
<code>string labels</code>	<code>true</code>	Show the string numbers 1–5.
<code>fret numbers</code>	<code>true</code>	Show the fret numbers under the board.
<code>markers</code>	<code>true</code>	Show the inlay dots (frets 5, 7, 10, 12, 15, 17, 19, 22, 24).
<code>focus</code>	<code>false</code>	Grey out the unpainted notes when something is painted.
<code>title</code>	<code>empty</code>	Title printed above the diagram.
<code>note font</code>	<code>\sffamily\bfseries</code>	Font commands for the note names.

Boolean options can be written without a value (`focus` means `focus=true`).

```
\begin{banjofret}[last fret=7, accidentals=flat, width=10cm]
  \bfpaintnote{Bb, Eb}{amber}
\end{banjofret}
```



```
\begin{banjofret}[last fret=7, left handed, string order=reverse, width=10cm]
  \bfpaintnote{G}{root}
\end{banjofret}
```



8 Tips and troubleshooting

- **Page size.** 24 frets with stacked note names are readable in landscape (`pdfscape`, or a landscape geometry). On portrait pages use a window such as `last fret=12`, or `accidentals=sharp`.
- **Font size warnings.** The note labels are scaled to the disc. With the default Computer Modern fonts in OT1 encoding $\text{\LaTeX}$ may warn that some sizes are unavailable. Loading `lmodern` (with `fontenc T1`) or any scalable font family removes them.
- **A # inside another macro.** In `\bfpaintnote` a literal # works in the document body, but inside macro definitions write `s` instead (`Fs`).
- **Errors.** Invalid positions, unknown note names and unknown options give clear error messages and the offending item is ignored.
- **Engines.** Tested with $\text{pdf}\text{\LaTeX}$, $\text{Xe}\text{\LaTeX}$ and $\text{Lua}\text{\LaTeX}$.
- **Combining with text.** The environment produces a centred block and can be put in a `figure` float.

9 Roadmap

Ideas for future versions: vertical (chord-box) orientation, fingering numbers inside the discs, scale-degree labels instead of note names, and ready-made scale and chord macros. Contributions are welcome.

10 Implementation

This section documents the source code. All internal functions and variables use the prefix `bfp` following the `expl3` conventions. Diagrams are drawn at the end of the environment, so that the drawing order is guaranteed: board and frets, then strings, and finally the coloured discs with their note names.

10.1 Source code

10.1.1 Package identification and dependencies

```

1 <*package>
2 <package>\NeedsTeXFormat{LaTeX2e}[2022-06-01]
3 \RequirePackage{expl3}
4 \ProvidesExplPackage{banjofret}{2026/09/28}{1.0.0}
5   {Five-string banjo fretboard diagrams with colour-coded chromatic notes}
6 \ifl@t@r\fmtversion{2022-06-01}{\}
7   {\PackageError{banjofret}{LaTeX 2022-06-01 or newer is required}{Please update your TeX distribution.}}
8 \ExplSyntaxOff
9 \RequirePackage{l3keys2e}
10 \RequirePackage{xcolor}
11 \RequirePackage{tikz}
12 \usetikzlibrary{calc}
13 \ExplSyntaxOn

```

10.1.2 Messages

```
14 \msg_new:nnn { banjofret } { unknown-key }
15 { Unknown~option~'#1'. }
16 \msg_new:nnnn { banjofret } { outside-env }
17 { #1~can~only~be~used~inside~a~'banjofret'~environment. }
18 { Put~the~command~between~begin~\{banjofret\}~and~end~\{banjofret\}. }
19 \msg_new:nnnn { banjofret } { bad-note }
20 { Unknown~note~name~'#1'. }
21 { Use~a~letter~A--G~optionally~followed~by~\iow_char:N\# ~or~s~(sharp)~or~b~(flat),~e.g.~C,~Fs,~Bb. }
22 \msg_new:nnnn { banjofret } { bad-position }
23 { Invalid~position~(#{1,2})~ignored. }
24 { String~must~be~1--5,~fret~0--24;~string~5~only~exists~from~fret~5~onwards. }
25 \msg_new:nnnn { banjofret } { bad-pair }
26 { Cannot~read~'#1'~as~(string,fret). }
27 { Write~positions~as~(string,fret),~separated~by~semicolons,~e.g.~(3,5);(2,1). }
28 \msg_new:nnnn { banjofret } { bad-range }
29 { Fret~window~must~satisfy~0<=first<=last<=24. }
30 { Falling~back~to~the~full~fretboard~(0--24). }
31 \msg_new:nnnn { banjofret } { bad-tuning }
32 { The~'tuning'~option~needs~exactly~five~notes~(strings~1--5). }
33 { Falling~back~to~the~standard~open-G~tuning~D,B,G,D,G. }
34 \msg_new:nnn { banjofret } { legend-key-only }
35 { Legend~entry~'#1'~has~no~label~and~was~ignored.~Use~colour=label. }
```

10.1.3 Colour palette

```
36 \definecolor{bfblue}{HTML}{7DB4E8}
37 \definecolor{bfgreen}{HTML}{86CFA3}
38 \definecolor{bfamber}{HTML}{F5C563}
39 \definecolor{bfcoral}{HTML}{F29E8E}
40 \definecolor{bfviolet}{HTML}{B9A3E3}
41 \definecolor{bfteal}{HTML}{7BCFCF}
42 \definecolor{bfgray}{HTML}{C9CED6}
43 \colorlet{bfroot}{bfcoral}
44 \colorlet{bfthird}{bfamber}
45 \colorlet{bffifth}{bfblue}
46 \colorlet{bfseventh}{bfviolet}
47 \colorlet{bfextension}{bfteal}
48 \colorlet{bftarget}{bfgreen}
49 \colorlet{bfpassing}{bfgray}
```

10.1.4 Variables and constants

```
50 \int_new:N \l__bfret_first_int
51 \int_new:N \l__bfret_last_int
52 \int_new:N \l__bfret_acc_int
53 \int_new:N \l__bfret_order_int
54 \int_new:N \l__bfret_dir_int
55 \int_new:N \l__bfret_xoff_int
56 \int_new:N \l__bfret_cols_int
57 \int_new:N \l__bfret_tmpa_int
58 \int_new:N \l__bfret_tmpc_int
59 \int_new:N \l__bfret_hash_int
60 \int_new:N \l__bfret_count_int
61 \bool_new:N \l__bfret_inenv_bool
62 \bool_new:N \l__bfret_open_bool
63 \bool_new:N \l__bfret_lefty_bool
64 \bool_new:N \l__bfret_slabs_bool
65 \bool_new:N \l__bfret_fnumbers_bool
66 \bool_new:N \l__bfret_markers_bool
67 \bool_new:N \l__bfret_focus_bool
68 \dim_new:N \l__bfret_w_dim
69 \dim_new:N \l__bfret_ss_dim
70 \dim_new:N \l__bfret_r_dim
71 \dim_new:N \l__bfret_d_dim
72 \dim_new:N \l__bfret_hh_dim
73 \dim_new:N \l__bfret_left_dim
74 \dim_new:N \l__bfret_right_dim
75 \dim_new:N \l__bfret_total_dim
76 \dim_new:N \l__bfret_margin_dim
77 \dim_new:N \l__bfret_fsa_dim
78 \dim_new:N \l__bfret_fsb_dim
79 \dim_new:N \l__bfret_xa_dim
80 \dim_new:N \l__bfret_ya_dim
81 \dim_new:N \l__bfret_xb_dim
```

```

82 \dim_new:N \l__bfret_yb_dim
83 \dim_new:N \l__bfret_tmpa_dim
84 \dim_new:N \l__bfret_tmppb_dim
85 \dim_new:N \l__bfret_ybot_dim
86 \tl_new:N \l__bfret_width_tl
87 \tl_new:N \l__bfret_title_tl
88 \tl_new:N \l__bfret_notefont_tl
89 \tl_new:N \l__bfret_tmppa_tl
90 \tl_new:N \l__bfret_key_tl
91 \tl_new:N \l__bfret_col_tl
92 \tl_new:N \l__bfret_draw_tl
93 \tl_new:N \l__bfret_txt_tl
94 \str_new:N \l__bfret_tmppa_str
95 \str_new:N \l__bfret_head_str
96 \str_new:N \l__bfret_orig_str
97 \str_new:N \l__bfret_tail_str
98 \clist_new:N \l__bfret_tuning_clist
99 \clist_new:N \l__bfret_names_clist
100 \seq_new:N \l__bfret_tuning_seq
101 \seq_new:N \l__bfret_legend_seq
102 \seq_new:N \l__bfret_tmppa_seq
103 \seq_new:N \l__bfret_tmppb_seq
104 \prop_new:N \l__bfret_paint_prop
105 \cs_new:Npn \__bfret_s: { \ensuremath { \sharp } }
106 \cs_new:Npn \__bfret_f: { \ensuremath { \flat } }
107 \seq_const_from_clist:Nn \c__bfret_sharp_seq
108 { C , C\__bfret_s: , D , D\__bfret_s: , E , F , F\__bfret_s: , G , G\__bfret_s: , A , A\__bfret_s: , B }
109 \seq_const_from_clist:Nn \c__bfret_flat_seq
110 { C , D\__bfret_f: , D , E\__bfret_f: , E , F , G\__bfret_f: , G , A\__bfret_f: , A , B\__bfret_f: , B }
111 \clist_const:Nn \c__bfret_natural_clist { 0 , 2 , 4 , 5 , 7 , 9 , 11 }
112 \clist_const:Nn \c__bfret_stringwidth_clist { 0.4pt , 0.55pt , 0.75pt , 1pt , 0.4pt }
113 \clist_const:Nn \c__bfret_markers_clist { 5 , 7 , 10 , 12 , 15 , 17 , 19 , 22 , 24 }

```

10.1.5 Options

```

114 \keys_define:nn { banjofret }
115 {
116   first~fret      .int_set:N = \l__bfret_first_int ,
117   first~fret      .initial:n = 0 ,
118   last~fret       .int_set:N = \l__bfret_last_int ,
119   last~fret       .initial:n = 24 ,
120   width           .tl_set:N = \l__bfret_width_tl ,
121   width           .initial:n = \linewidth ,
122   accidentals     .choices:nn = { sharp , flat , both }
123   { \int_set:Nn \l__bfret_acc_int { \l_keys_choice_int } } ,
124   accidentals     .initial:n = both ,
125   tuning          .clist_set:N = \l__bfret_tuning_clist ,
126   tuning          .initial:n = { D , B , G , D , G } ,
127   string~order    .choices:nn = { tab , reverse }
128   { \int_set:Nn \l__bfret_order_int { 3 - 2 * \l_keys_choice_int } } ,
129   string~order    .initial:n = tab ,
130   left~handed     .bool_set:N = \l__bfret_lefty_bool ,
131   left~handed     .initial:n = false ,
132   string~labels   .bool_set:N = \l__bfret_slabs_bool ,
133   string~labels   .initial:n = true ,
134   fret~numbers    .bool_set:N = \l__bfret_fnumbers_bool ,
135   fret~numbers    .initial:n = true ,
136   markers         .bool_set:N = \l__bfret_markers_bool ,
137   markers         .initial:n = true ,
138   focus          .bool_set:N = \l__bfret_focus_bool ,
139   focus          .initial:n = false ,
140   title          .tl_set:N = \l__bfret_title_tl ,
141   title          .initial:n = { } ,
142   note~font      .tl_set:N = \l__bfret_notefont_tl ,
143   note~font      .initial:n = { \sfamily \bfseries } ,
144   unknown        .code:n =
145   { \msg_error:nnx { banjofret } { unknown-key } { \l_keys_key_str } }
146 }
147 \ProcessKeysOptions { banjofret }
148 \NewDocumentCommand \banjofretset { m }
149 { \keys_set:nn { banjofret } {#1} }

```

10.1.6 Note names and pitch classes

```

150 \cs_new_protected:Npn \__bfret_note_to_pc:nN #1#2
151 {

```

```

152 \str_set:Nx \l__bfret_tmpa_str { \tl_to_str:n {#1} }
153 \str_remove_all:Nn \l__bfret_tmpa_str { ~ }
154 \str_set_eq:NN \l__bfret_orig_str \l__bfret_tmpa_str
155 \str_set:Nx \l__bfret_head_str { \str_head:N \l__bfret_tmpa_str }
156 \str_set:Nx \l__bfret_tail_str { \str_tail:N \l__bfret_tmpa_str }
157 \exp_args:NV \str_case:nnF \l__bfret_head_str
158 {
159   { C } { \int_set:Nn #2 { 0 } }
160   { D } { \int_set:Nn #2 { 2 } }
161   { E } { \int_set:Nn #2 { 4 } }
162   { F } { \int_set:Nn #2 { 5 } }
163   { G } { \int_set:Nn #2 { 7 } }
164   { A } { \int_set:Nn #2 { 9 } }
165   { B } { \int_set:Nn #2 { 11 } }
166 }
167 {
168   \msg_error:nnV { banjofret } { bad-note } \l__bfret_orig_str
169   \int_set:Nn #2 { 0 }
170 }
171 \int_zero:N \l__bfret_hash_int
172 \str_map_inline:Nn \l__bfret_tail_str
173 {
174   \str_case:nnF {##1}
175   {
176     { s } { \int_incr:N #2 }
177     { b } { \int_decr:N #2 }
178   }
179   {
180     \str_if_eq:nVTF {##1} \c_hash_str
181     { \int_incr:N \l__bfret_hash_int }
182     { \msg_error:nnV { banjofret } { bad-note } \l__bfret_orig_str }
183   }
184 }
185 % A '#' character is doubled by \tl_to_str:n, hence the division by two.
186 \int_add:Nn #2 { \int_div_truncate:nn { \l__bfret_hash_int } { 2 } }
187 \int_set:Nn #2 { \int_mod:nn { \int_mod:nn {#2} { 12 } + 12 } { 12 } }
188 }
189 \cs_new:Npn \__bfret_pc:nn #1#2
190 {
191   \int_mod:nn
192   { \seq_item:Nn \l__bfret_tuning_seq {#1} + #2 - \int_compare:nTF { #1 = 5 } { 5 } { 0 } }
193   { 12 }
194 }
195 \cs_new_protected:Npn \__bfret_parse_tuning:
196 {
197   \seq_clear:N \l__bfret_tuning_seq
198   \clist_map_inline:Nn \l__bfret_tuning_clist
199   {
200     \__bfret_note_to_pc:nN {##1} \l__bfret_tmpc_int
201     \seq_put_right:NV \l__bfret_tuning_seq \l__bfret_tmpc_int
202   }
203   \int_compare:nF { \seq_count:N \l__bfret_tuning_seq = 5 }
204   {
205     \msg_error:nn { banjofret } { bad-tuning }
206     \seq_set_split:Nnn \l__bfret_tuning_seq { , } { 2 , 11 , 7 , 2 , 7 }
207   }
208 }

```

10.1.7 Colour resolution

```

209 \cs_new:Npn \__bfret_color:n #1
210 {
211   \cs_if_exist:cTF { \c_backslash_str color@bf#1 } { bf#1 } {#1}
212 }

```

10.1.8 Painting commands

```

213 \prg_new_conditional:Npnn \__bfret_valid:nn #1#2 { p , T , F , TF }
214 {
215   \bool_if:nTF
216   {
217     \int_compare_p:nNn {#1} > { 0 } && \int_compare_p:nNn {#1} < { 6 } &&
218     \int_compare_p:nNn {#2} > { -1 } && \int_compare_p:nNn {#2} < { 25 } &&
219     ( \int_compare_p:nNn {#1} < { 5 } || \int_compare_p:nNn {#2} > { 4 } )
220   }
221   { \prg_return_true: }

```

```

222     { \prg_return_false: }
223 }
224 \cs_new_protected:Npn \__bfret_check_env:n #1
225 {
226     \bool_if:NF \l__bfret_inenv_bool
227     { \msg_error:nnn { banjofret } { outside-env } {#1} }
228 }
229 \cs_new_protected:Npn \__bfret_put:nnn #1#2#3
230 {
231     \__bfret_valid:nnT {#1} {#2}
232     {
233         \tl_set:Nx \l__bfret_key_tl { \int_eval:n {#1} , \int_eval:n {#2} }
234         \prop_put:NVn \l__bfret_paint_prop \l__bfret_key_tl {#3}
235     }
236 }
237 \cs_new_protected:Npn \__bfret_paint_pair:nn #1#2
238 {
239     \tl_if_blank:nF {#1}
240     {
241         \tl_set:Nn \l__bfret_tmpa_tl {#1}
242         \tl_remove_all:Nn \l__bfret_tmpa_tl { ( }
243         \tl_remove_all:Nn \l__bfret_tmpa_tl { ) }
244         \seq_set_split:NnV \l__bfret_tmpb_seq { , } \l__bfret_tmpa_tl
245         \int_compare:nTF { \seq_count:N \l__bfret_tmpb_seq = 2 }
246         {
247             \__bfret_valid:nnTF
248             { \seq_item:Nn \l__bfret_tmpb_seq { 1 } }
249             { \seq_item:Nn \l__bfret_tmpb_seq { 2 } }
250             {
251                 \__bfret_put:nnn
252                 { \seq_item:Nn \l__bfret_tmpb_seq { 1 } }
253                 { \seq_item:Nn \l__bfret_tmpb_seq { 2 } }
254                 {#2}
255             }
256             {
257                 \msg_error:nnxx { banjofret } { bad-position }
258                 { \seq_item:Nn \l__bfret_tmpb_seq { 1 } }
259                 { \seq_item:Nn \l__bfret_tmpb_seq { 2 } }
260             }
261         }
262         { \msg_error:nnn { banjofret } { bad-pair } {#1} }
263     }
264 }
265 \NewDocumentCommand \bfpaint { m m }
266 {
267     \__bfret_check_env:n { \bfpaint }
268     \seq_set_split:Nnn \l__bfret_tmpa_seq { ; } {#1}
269     \seq_map_inline:Nn \l__bfret_tmpa_seq
270     { \__bfret_paint_pair:nn {##1} {#2} }
271 }
272 \cs_new_protected:Npn \__bfret_range:nNN #1#2#3
273 {
274     \seq_set_split:Nnn \l__bfret_tmpb_seq { - } {#1}
275     \int_set:Nn #2 { \seq_item:Nn \l__bfret_tmpb_seq { 1 } }
276     \int_compare:nTF { \seq_count:N \l__bfret_tmpb_seq > 1 }
277     { \int_set:Nn #3 { \seq_item:Nn \l__bfret_tmpb_seq { 2 } } }
278     { \int_set_eq:NN #3 #2 }
279 }
280 \int_new:N \l__bfret_ra_int
281 \int_new:N \l__bfret_rb_int
282 \NewDocumentCommand \bfpaintstring { m m m }
283 {
284     \__bfret_check_env:n { \bfpaintstring }
285     \__bfret_range:nNN {#2} \l__bfret_ra_int \l__bfret_rb_int
286     \int_step_inline:nnn { \l__bfret_ra_int } { \l__bfret_rb_int }
287     { \__bfret_put:nnn {#1} {##1} {#3} }
288 }
289 \NewDocumentCommand \bfpaintfret { m m }
290 {
291     \__bfret_check_env:n { \bfpaintfret }
292     \__bfret_range:nNN {#1} \l__bfret_ra_int \l__bfret_rb_int
293     \int_step_inline:nnn { \l__bfret_ra_int } { \l__bfret_rb_int }
294     {
295         \int_step_inline:nn { 5 }
296         { \__bfret_put:nnn { ####1 } { ##1 } {#2} }

```

```

297     }
298 }
299 \cs_new_protected:Npn \__bfret_paint_pc_string:nnn #1#2#3
300 {
301     \int_step_inline:nnn { 0 } { 24 }
302     {
303         \int_compare:nT { \__bfret_pc:nn {#3} { ##1 } = #1 }
304         { \__bfret_put:nnn {#3} { ##1 } {#2} }
305     }
306 }
307 \NewDocumentCommand \bfpaintnote { m m }
308 {
309     \__bfret_check_env:n { \bfpaintnote }
310     % Convert to a string first so that a literal '#' in the input is harmless.
311     \clist_set:Nx \l__bfret_names_clist { \tl_to_str:n {#1} }
312     \clist_map_inline:Nn \l__bfret_names_clist
313     {
314         \__bfret_note_to_pc:nN { ##1 } \l__bfret_tmpc_int
315         \int_step_inline:nn { 5 }
316         {
317             \__bfret_paint_pc_string:nnn
318             { \int_use:N \l__bfret_tmpc_int } {#2} { #####1 }
319         }
320     }
321 }
322 \cs_new_protected:Npn \__bfret_legend_key:n #1
323 { \msg_warning:nnn { banjofret } { legend-key-only } {#1} }
324 \cs_new_protected:Npn \__bfret_legend_pair:nn #1#2
325 { \seq_put_right:Nn \l__bfret_legend_seq { { #1 } { #2 } } }
326 \NewDocumentCommand \bflegend { m }
327 {
328     \__bfret_check_env:n { \bflegend }
329     \seq_clear:N \l__bfret_legend_seq
330     \keyval_parse:NnN \__bfret_legend_key:n \__bfret_legend_pair:nn {#1}
331 }

```

10.1.9 Geometry helpers

```

332 \cs_new:Npn \__bfret_xpos:n #1
333 {
334     \dim_eval:n
335     {
336         ( \int_eval:n { #1 - \l__bfret_xoff_int } \l__bfret_w_dim - 0.5 \l__bfret_w_dim )
337         * \l__bfret_dir_int
338     }
339 }
340 \cs_new:Npn \__bfret_ypos:n #1
341 { \dim_eval:n { \int_eval:n { \l__bfret_order_int * ( 3 - #1 ) } \l__bfret_ss_dim } }
342 \cs_new_protected:Npn \__bfret_seg:nnnnn #1#2#3#4#5
343 {
344     \dim_set:Nn \l__bfret_xa_dim { ( #1 ) * \l__bfret_dir_int }
345     \dim_set:Nn \l__bfret_ya_dim { #2 }
346     \dim_set:Nn \l__bfret_xb_dim { ( #3 ) * \l__bfret_dir_int }
347     \dim_set:Nn \l__bfret_yb_dim { #4 }
348     \draw [ #5 ]
349         ( \l__bfret_xa_dim , \l__bfret_ya_dim ) --
350         ( \l__bfret_xb_dim , \l__bfret_yb_dim ) ;
351 }
352 \cs_new_protected:Npn \__bfret_fs:n #1
353 { \exp_args:Nx \__bfret_fs_aux:n { \dim_to_decimal_in_unit:nn {#1} { 1pt } } }
354 \cs_new_protected:Npn \__bfret_fs_aux:n #1
355 { \fontsize {#1} {#1} \selectfont }
356 \cs_new_protected:Npn \__bfret_setup_geometry:
357 {
358     \bool_if:nF
359     {
360         \int_compare_p:nNn { \l__bfret_first_int } > { -1 } &&
361         \int_compare_p:nNn { \l__bfret_first_int } < { 25 } &&
362         \int_compare_p:nNn { \l__bfret_last_int } > { \l__bfret_first_int - 1 } &&
363         \int_compare_p:nNn { \l__bfret_last_int } < { 25 }
364     }
365     {
366         \msg_error:nn { banjofret } { bad-range }
367         \int_set:Nn \l__bfret_first_int { 0 }
368         \int_set:Nn \l__bfret_last_int { 24 }

```

```

369     }
370     \int_set:Nn \l__bfret_xoff_int { \int_max:nn { \l__bfret_first_int - 1 } { 0 } }
371     \bool_set:Nn \l__bfret_open_bool
372     { \int_compare_p:nNn { \l__bfret_first_int } = { 0 } }
373     \int_set:Nn \l__bfret_cols_int
374     { \l__bfret_last_int - \l__bfret_xoff_int + \bool_if:NTF \l__bfret_open_bool { 1 } { 0 } }
375     \int_set:Nn \l__bfret_dir_int { \bool_if:NTF \l__bfret_lefty_bool { -1 } { 1 } }
376     \dim_set:Nn \l__bfret_total_dim { \l__bfret_width_tl }
377     \dim_set:Nn \l__bfret_margin_dim
378     { \bool_if:NTF \l__bfret_slablabels_bool { 14pt } { 0pt } }
379     \dim_set:Nn \l__bfret_w_dim
380     { ( \l__bfret_total_dim - \l__bfret_margin_dim ) / \l__bfret_cols_int }
381     \dim_set:Nn \l__bfret_ss_dim
382     { \dim_min:nn { 0.8 \l__bfret_w_dim } { 26pt } }
383     \dim_set:Nn \l__bfret_r_dim
384     { \dim_min:nn { \l__bfret_w_dim } { \l__bfret_ss_dim } }
385     \dim_set:Nn \l__bfret_r_dim { 0.44 \l__bfret_r_dim }
386     \dim_set:Nn \l__bfret_d_dim { 2 \l__bfret_r_dim }
387     \dim_set:Nn \l__bfret_fsa_dim { 1.15 \l__bfret_r_dim }
388     \dim_set:Nn \l__bfret_fsb_dim { 0.8 \l__bfret_r_dim }
389     \dim_set:Nn \l__bfret_hh_dim { 2.6 \l__bfret_ss_dim }
390     \dim_set:Nn \l__bfret_left_dim
391     { \bool_if:NTF \l__bfret_open_bool { - \l__bfret_w_dim } { 0pt } }
392     \dim_set:Nn \l__bfret_right_dim
393     { \int_eval:n { \l__bfret_last_int - \l__bfret_xoff_int } \l__bfret_w_dim }
394 }

```

10.1.10 Drawing the fretboard

```

395 \cs_new_protected:Npn \__bfret_draw_fretlines:
396 {
397     \int_step_inline:nnn { \l__bfret_xoff_int } { \l__bfret_last_int }
398     {
399         \dim_set:Nn \l__bfret_tmpa_dim
400         { \int_eval:n { ##1 - \l__bfret_xoff_int } \l__bfret_w_dim }
401         \bool_if:nTF
402         { \int_compare_p:nNn {##1} = { \l__bfret_xoff_int } && \bool_if_p:N \l__bfret_open_bool }
403         {
404             \__bfret_seg:nnnnn
405             { \l__bfret_tmpa_dim } { - \l__bfret_hh_dim }
406             { \l__bfret_tmpa_dim } { \l__bfret_hh_dim }
407             { line~width = 3pt , line~cap = butt }
408         }
409         {
410             \__bfret_seg:nnnnn
411             { \l__bfret_tmpa_dim } { - \l__bfret_hh_dim }
412             { \l__bfret_tmpa_dim } { \l__bfret_hh_dim }
413             { line~width = 0.6pt }
414         }
415     }
416 }
417 \cs_new_protected:Npn \__bfret_draw_strings:
418 {
419     \int_step_inline:nn { 5 }
420     {
421         \bool_if:nT
422         { \int_compare_p:nNn {##1} < { 5 } || \int_compare_p:nNn { \l__bfret_last_int } > { 4 } }
423         {
424             \dim_set:Nn \l__bfret_tmpb_dim
425             {
426                 \bool_if:nTF
427                 { \int_compare_p:nNn {##1} = { 5 } && \int_compare_p:nNn { \l__bfret_first_int } < { 6 } }
428                 { \int_eval:n { 5 - \l__bfret_xoff_int } \l__bfret_w_dim - 0.5 \l__bfret_w_dim }
429                 { \l__bfret_left_dim }
430             }
431             \__bfret_seg:nnnnn
432             { \l__bfret_tmpb_dim } { \__bfret_ypos:n {##1} }
433             { \l__bfret_right_dim } { \__bfret_ypos:n {##1} }
434             { line~width = \clist_item:Nn \c__bfret_stringwidth_clist {##1} }
435         }
436     }
437 }
438 \cs_new_protected:Npn \__bfret_label:
439 {
440     \tl_use:N \l__bfret_notefont_tl

```

```

441 \clist_if_in:NVTF \c__bfret_natural_clist \l__bfret_tmpc_int
442 {
443   \__bfret_fs:n { \l__bfret_fsa_dim }
444   \seq_item:Nn \c__bfret_sharp_seq { \l__bfret_tmpc_int + 1 }
445 }
446 {
447   \int_case:nn { \l__bfret_acc_int }
448   {
449     { 1 }
450     {
451       \__bfret_fs:n { \l__bfret_fsa_dim }
452       \seq_item:Nn \c__bfret_sharp_seq { \l__bfret_tmpc_int + 1 }
453     }
454     { 2 }
455     {
456       \__bfret_fs:n { \l__bfret_fsa_dim }
457       \seq_item:Nn \c__bfret_flat_seq { \l__bfret_tmpc_int + 1 }
458     }
459     { 3 }
460     {
461       \__bfret_fs:n { \l__bfret_fsb_dim }
462       \shortstack [ c ]
463       {
464         \seq_item:Nn \c__bfret_sharp_seq { \l__bfret_tmpc_int + 1 } \
465         \seq_item:Nn \c__bfret_flat_seq { \l__bfret_tmpc_int + 1 }
466       }
467     }
468   }
469 }
470 }
471 \cs_new_protected:Npn \__bfret_disc:nn #1#2
472 {
473   \__bfret_valid:nnT {#1} {#2}
474   {
475     \dim_set:Nn \l__bfret_xa_dim { \__bfret_xpos:n {#2} }
476     \dim_set:Nn \l__bfret_ya_dim { \__bfret_ypos:n {#1} }
477     \int_set:Nn \l__bfret_tmpc_int { \__bfret_pc:nn {#1} {#2} }
478     \prop_get:NnNTF \l__bfret_paint_prop { #1 , #2 } \l__bfret_tmpa_tl
479     {
480       \tl_set:Nx \l__bfret_col_tl { \__bfret_color:n { \l__bfret_tmpa_tl } }
481       \node
482       [ circle , draw = black , fill = \l__bfret_col_tl , line~width = 0.6pt ,
483         inner~sep = 0pt , minimum~size = \l__bfret_d_dim , text = black ]
484       at ( \l__bfret_xa_dim , \l__bfret_ya_dim )
485       { \__bfret_label: } ;
486     }
487     {
488       \bool_lazy_and:nnTF
489       { \bool_if_p:N \l__bfret_focus_bool }
490       { ! \prop_if_empty_p:N \l__bfret_paint_prop }
491       {
492         \node
493         [ circle , draw = black!30 , fill = white , line~width = 0.5pt ,
494           inner~sep = 0pt , minimum~size = \l__bfret_d_dim , text = black!40 ]
495         at ( \l__bfret_xa_dim , \l__bfret_ya_dim )
496         { \__bfret_label: } ;
497       }
498       {
499         \node
500         [ circle , draw = black , fill = white , line~width = 0.5pt ,
501           inner~sep = 0pt , minimum~size = \l__bfret_d_dim , text = black ]
502         at ( \l__bfret_xa_dim , \l__bfret_ya_dim )
503         { \__bfret_label: } ;
504       }
505     }
506   }
507 }
508 \cs_new_protected:Npn \__bfret_draw_discs:
509 {
510   \int_step_inline:nn { 5 }
511   {
512     \int_step_inline:nnn { \l__bfret_first_int } { \l__bfret_last_int }
513     { \__bfret_disc:nn { ##1 } { #####1 } }
514   }
515 }

```

```

516 \cs_new_protected:Npn \__bfret_draw_labels:
517 {
518   \bool_if:NT \l__bfret_slab_labels_bool
519   {
520     \int_step_inline:nn { 5 }
521     {
522       \dim_set:Nn \l__bfret_xa_dim
523       { ( \l__bfret_left_dim - 4pt ) * \l__bfret_dir_int }
524       \dim_set:Nn \l__bfret_ya_dim { \__bfret_ypos:n {##1} }
525       \node
526       [ anchor = \bool_if:NTF \l__bfret_lefty_bool { west } { east } ,
527         inner~sep = 0pt , text = black ]
528       at ( \l__bfret_xa_dim , \l__bfret_ya_dim )
529       { \sffamily \footnotesize ##1 } ;
530     }
531   }
532 }
533 \cs_new_protected:Npn \__bfret_draw_frettrack:
534 {
535   \dim_set:Nn \l__bfret_ybot_dim { - \l__bfret_hh_dim }
536   \bool_if:NT \l__bfret_fnumbers_bool
537   {
538     \dim_sub:Nn \l__bfret_ybot_dim { 9pt }
539     \int_step_inline:nnn { \int_max:nn { \l__bfret_first_int } { 1 } } { \l__bfret_last_int }
540     {
541       \dim_set:Nn \l__bfret_xa_dim { \__bfret_xpos:n {##1} }
542       \node [ inner~sep = 0pt , text = black ]
543       at ( \l__bfret_xa_dim , \l__bfret_ybot_dim )
544       { \sffamily \footnotesize ##1 } ;
545     }
546   }
547   \bool_if:NT \l__bfret_markers_bool
548   {
549     \dim_sub:Nn \l__bfret_ybot_dim { 9pt }
550     \clist_map_inline:Nn \c__bfret_markers_clist
551     {
552       \bool_if:NNT
553       {
554         \int_compare_p:nNn {##1} > { \l__bfret_first_int - 1 } &&
555         \int_compare_p:nNn {##1} < { \l__bfret_last_int + 1 }
556       }
557       {
558         \dim_set:Nn \l__bfret_xa_dim { \__bfret_xpos:n {##1} }
559         \int_compare:NTF { \int_mod:nn {##1} { 12 } = 0 }
560         {
561           \dim_set:Nn \l__bfret_xb_dim { \l__bfret_xa_dim - 3pt }
562           \fill ( \l__bfret_xb_dim , \l__bfret_ybot_dim ) circle ( 1.8pt ) ;
563           \dim_set:Nn \l__bfret_xb_dim { \l__bfret_xa_dim + 3pt }
564           \fill ( \l__bfret_xb_dim , \l__bfret_ybot_dim ) circle ( 1.8pt ) ;
565         }
566         { \fill ( \l__bfret_xa_dim , \l__bfret_ybot_dim ) circle ( 1.8pt ) ; }
567       }
568     }
569   }
570 }
571 \cs_new_protected:Npn \__bfret_legend_item:nn #1#2
572 {
573   \tl_set:Nx \l__bfret_col_tl { \__bfret_color:n {#1} }
574   \node
575   [ circle , draw = black , fill = \l__bfret_col_tl , line~width = 0.6pt ,
576     inner~sep = 0pt , minimum~size = 10pt , anchor = west ]
577   ( bfret-lc ) at ( bfret-legpos ) { } ;
578   \node
579   [ anchor = west , inner~xsep = 4pt , inner~ysep = 0pt , text = black ]
580   ( bfret-lt ) at ( bfret-lc.east ) { \sffamily \footnotesize #2 } ;
581   \coordinate ( bfret-legpos ) at ( $ ( bfret-lt.east ) + ( 12pt , 0pt ) $ ) ;
582 }
583 \cs_new_protected:Npn \__bfret_draw_legend:
584 {
585   \seq_if_empty:NF \l__bfret_legend_seq
586   {
587     \dim_sub:Nn \l__bfret_ybot_dim { 14pt }
588     \dim_set:Nn \l__bfret_xa_dim
589     { \bool_if:NTF \l__bfret_lefty_bool { - \l__bfret_right_dim } { \l__bfret_left_dim } }
590     \coordinate ( bfret-legpos ) at ( \l__bfret_xa_dim , \l__bfret_ybot_dim ) ;

```

```

591     \seq_map_inline:Nn \l__bfret_legend_seq { \__bfret_legend_item:nn ##1 }
592   }
593 }
594 \cs_new_protected:Npn \__bfret_draw_title:
595 {
596   \tl_if_blank:VF \l__bfret_title_tl
597   {
598     \dim_set:Nn \l__bfret_xa_dim
599     { ( \l__bfret_left_dim + \l__bfret_right_dim ) / 2 * \l__bfret_dir_int }
600     \dim_set:Nn \l__bfret_ya_dim { \l__bfret_hh_dim + 6pt }
601     \node [ anchor = south , inner~sep = 0pt , text = black ]
602       at ( \l__bfret_xa_dim , \l__bfret_ya_dim )
603       { \sffamily \bfseries \large \l__bfret_title_tl } ;
604   }
605 }

```

10.1.11 The banjofret environment

```

606 \cs_new_protected:Npn \__bfret_begin:n #1
607 {
608   \prop_clear:N \l__bfret_paint_prop
609   \seq_clear:N \l__bfret_legend_seq
610   \bool_set_true:N \l__bfret_inenv_bool
611   \keys_set:nn { banjofret } {#1}
612   \__bfret_parse_tuning:
613   \__bfret_setup_geometry:
614 }
615 \cs_new_protected:Npn \__bfret_end:
616 {
617   \par \medskip \noindent \hfil
618   \begin{tikzpicture}
619     \__bfret_draw_title:
620     \__bfret_seg:nnnnn
621     { \l__bfret_left_dim } { \l__bfret_hh_dim }
622     { \l__bfret_right_dim } { \l__bfret_hh_dim } { line~width = 0.8pt }
623     \__bfret_seg:nnnnn
624     { \l__bfret_left_dim } { - \l__bfret_hh_dim }
625     { \l__bfret_right_dim } { - \l__bfret_hh_dim } { line~width = 0.8pt }
626     \__bfret_draw_fretlines:
627     \__bfret_draw_strings:
628     \__bfret_draw_discs:
629     \__bfret_draw_labels:
630     \__bfret_draw_frettrack:
631     \__bfret_draw_legend:
632   \end{tikzpicture}
633   \hfil \par \medskip
634 }
635 \NewDocumentEnvironment { banjofret } { 0{} }
636 { \__bfret_begin:n {#1} }
637 { \__bfret_end: }
638 \end{package}

```